

vertex = v
edge = e

Unit (1)
Graph Theory

M-2110

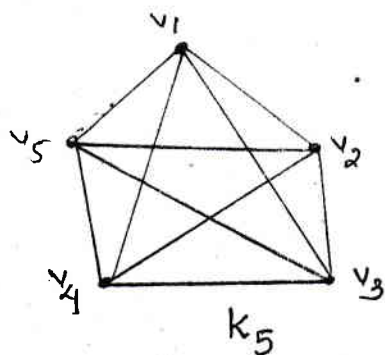
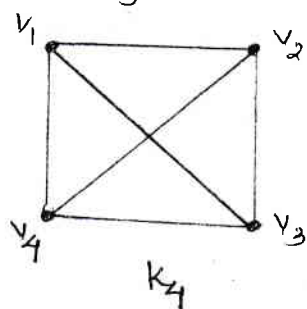
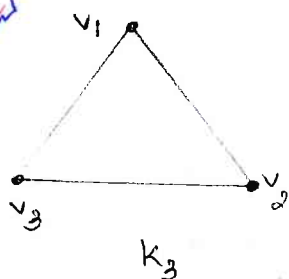
Definition

A graph (or undirected graph) G consists of a set V of vertices (or nodes) and a set E of edges (or arcs) such that each edge $e \in E$ is associated with an unordered pair of vertices. If there is a unique edge e associated with the vertices v and w , we write $e = (v, w)$ or $e = (w, v)$.

Definition

The graph K_n called the complete graph on n vertices, has n vertices and every vertex is joined to every other vertex by an edge.

① Draw K_3 , K_4 and K_5 .



$$5C_2 = \frac{5 \times 4}{1 \times 2} = 10$$

② Find a formula for the number of edges in K_n .
solution,

The graph K_n is a complete graph with n vertices. It has no loop and no parallel edges. So it is a simple graph. There is at most one edge between ~~two distinct~~ two distinct vertices.

$$\begin{aligned} \text{The number of edges} &= {}^nC_2 \\ &= \frac{n(n-1)}{2} \text{ edges} \end{aligned}$$

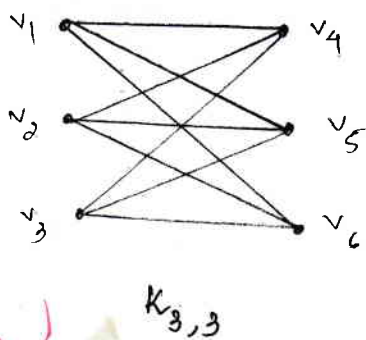
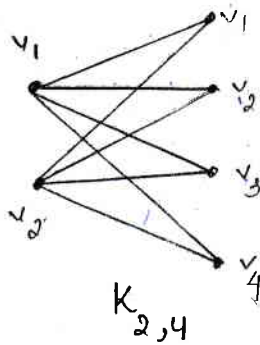
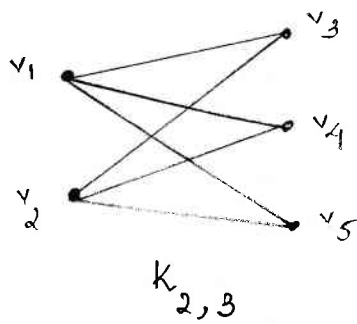
Definition

A graph in which the vertices can be partitioned into disjoint sets V_1 and V_2 with every edge incident on one vertex in V_1 and one vertex in V_2 is called a bipartite graph.

The graph $K_{m,n}$ called the complete bipartite graph on m and n vertices has disjoint sets V_1 of m vertices and V_2 of n vertices. Every vertex in V_1 is joined to every vertex in V_2 by an edge.

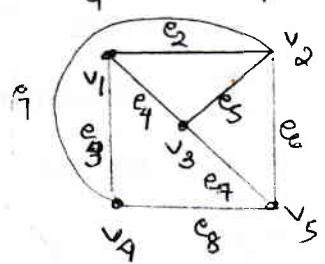
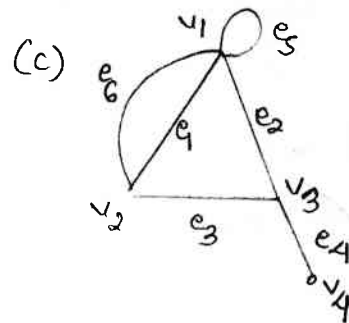
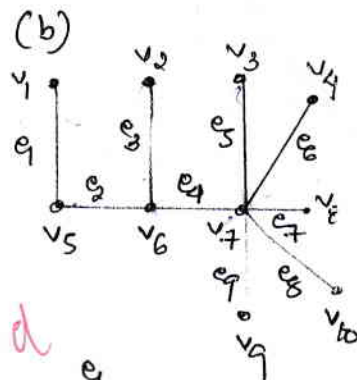
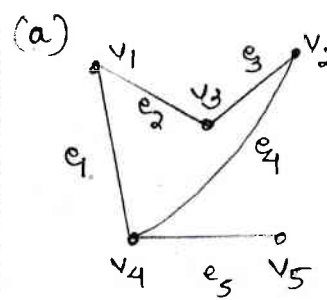
For question 2 (bipartite G)

② Draw $K_{2,3}$, $K_{2,4}$ and $K_{3,3}$.

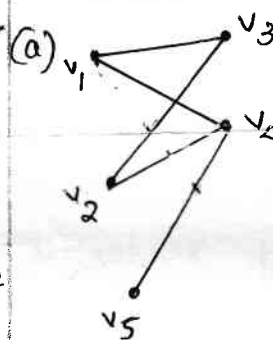


For

④ IF the graph is bipartite specify the disjoint vertex sets.



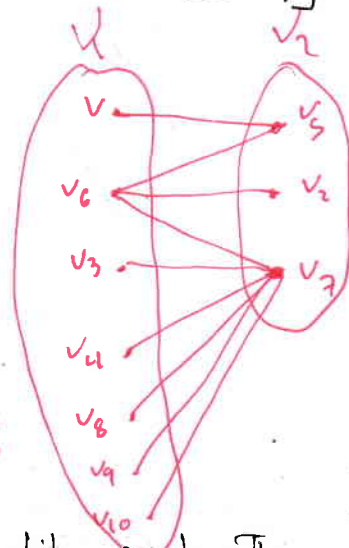
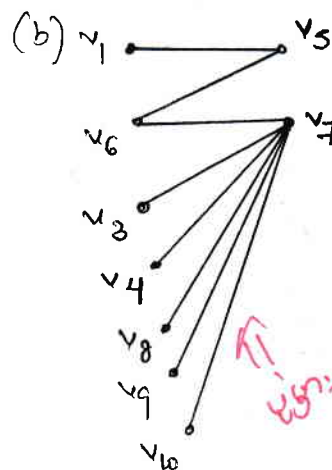
solution



The graph is bipartite graph.

The disjoint vertex sets

$$V_1 = \{v_1, v_2, v_5\} \text{ and } V_2 = \{v_3, v_4\}$$



The graph is bipartite graph. The disjoint vertex sets $V_1 = \{v_1, v_3, v_4, v_6, v_8, v_9, v_{10}\}$, $V_2 = \{v_5, v_7\}$.

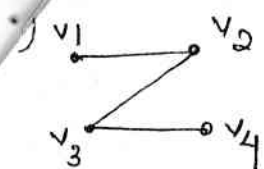
③ Find a formula for the number of $K_{m,n}$.

solution,

By definition, a complete bipartite graph $K_{m,n}$ divides its vertices into two disjoint sets, V_1 and V_2 .

The vertex set V_1 contains m vertices and the vertex set V_2 contain n vertices.

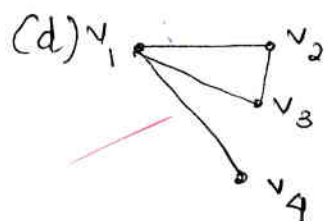
The total number of edges in $K_{m,n}$ is $m \times n = mn$ edges.



The graph G is bipartite graph.

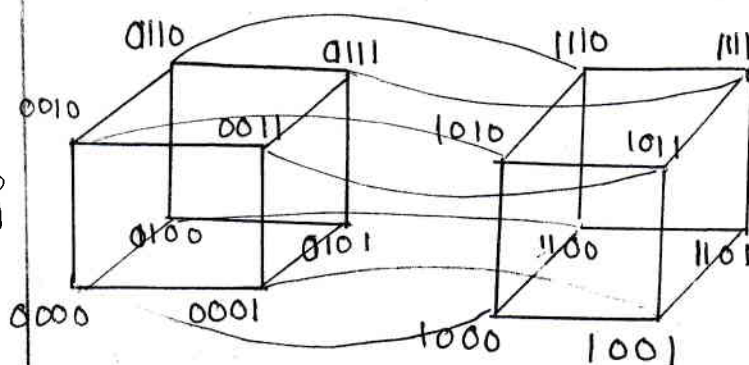
The disjoint vertex sets $V_1 = \{v_1, v_3\}$

and $V_2 = \{v_2, v_4\}$.



The graph G is not bipartite graph.

n cube



The number of vertices in n -cube $= 2^n$ vertices.

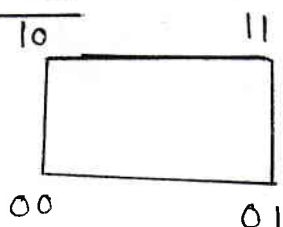
The number of edges in n -cube $= n \cdot 2^{n-1}$ edges

(5) The n -cube (Hypercube)

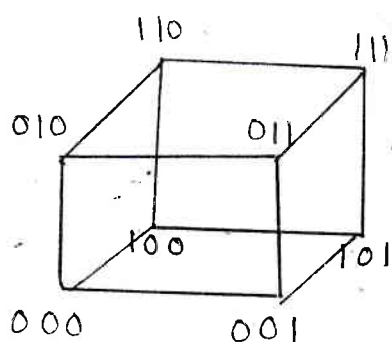
1 cube



2 cube



3 cube

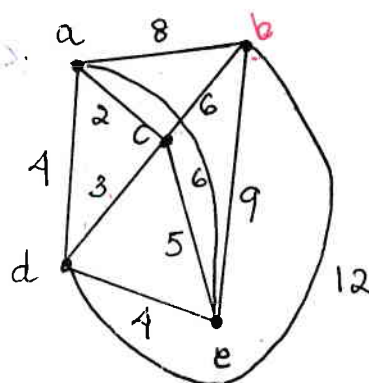


(6) Find a path of minimum length from v to w in the graph of figure that passes through each vertex exactly one time.

(i) $v = a, w = e$

(ii) $v = c, w = d$

(iii) $v = a, w = b$.



(a) solution,

path	length
a, b, c, d, e	$8+6+3+4=21$
a, b, d, c, e	$8+12+3+5=28$
a, c, b, d, e	$2+6+12+4=24$
a, c, d, b, e	$2+3+12+9=26$
a, d, b, c, e	$4+12+6+5=27$
a, d, e, b, c	$4+3+6+9=22$

The path of minimum length from a to e is a, b, c, d, e with length of 21.

(b) $v=c, w=d$

path	length
c, a, b, e, d	$2+8+9+4=23$
c, a, e, b, d	$2+6+9+12=29$
c, b, a, e, d	$5+8+6+4=23$
e, b, e, a, d	$6+9+6+4=25$
c, e, a, b, d	$5+6+8+12=31$
c, e, b, a, d	$5+9+8+4=26$

The path of minimum length from c to d is c, a, b, e, d with length of 23.

(c) $v=a, w=b$

path	length
a, c, d, e, b	
a, d, e, b	
a, e, c, d, b	
a, c, e, d, b	
a, c, d, e, b	
a, d, e, c, b	

The path of minimum length from a to b is with length of .

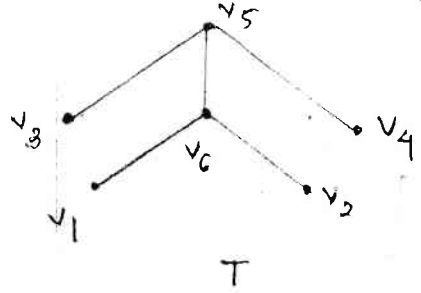
Draw a graph having the given properties or explain why no such graph exists

- (i) Tree; all vertices of degree 2.
- (ii) Tree; six vertices having degrees 1, 1, 1, 1, 3, 3.
- (iii) Six edges; eight vertices
- (iv) Acyclic; four edges; six vertices
- (v) Tree; four internal vertices, six terminal vertices.

solution,

(i) Tree; all vertices of degree 2.
A terminal vertex has degree 1.
No such graph.

(ii) Tree; six vertices having 1, 1, 1, 1, 3, 3.



(iii)

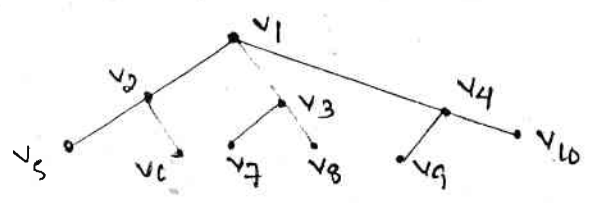
$$2e \geq 4f$$

$$2e \geq 4(e - v + 2)$$

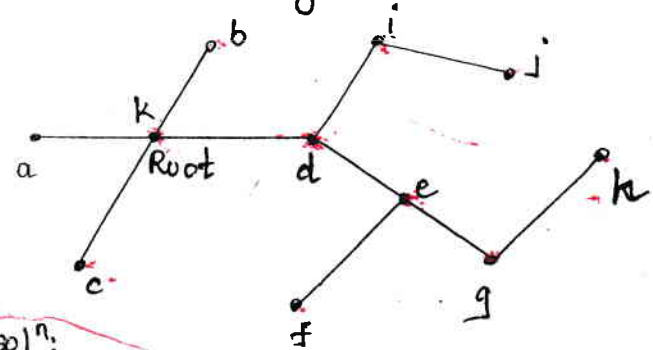
$$2(9) \geq 4(9 - 6 + 2)$$

$$18 \geq 20$$

(v) Tree; four internal vertices, six terminal vertices



② Find the level of each vertex and the height of the tree.



solⁿ:

The vertices a, b, c, d, e, f, g, h, i, j, k are on (respectively) levels 1, 1, 1, 1, 2, 3, 3, 4, 2, 3, 0. The height of T is 4.

③ Show that the graph $K_{3,3}$ is not planar.

Proof: Suppose that $K_{3,3}$ is planar. Since every cycle has at least four edges, each face is bounded by at least four edges. Thus the number of edges that bound faces is at least $4f$. In a planar graph, each edge belongs to at most two bounding cycles. Therefore,



$$2e \geq 4f$$

$$2e \geq 4(e - v + 2)$$

$$e = 9, v = 6, 18 \geq 2$$

$$9 \geq 4(9 - 6 + 2) = 20$$

which is contradiction. Therefore, $K_{3,3}$ is not planar.

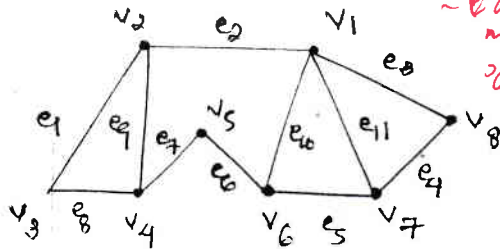
face in a planar graph $(e - v + 2)$

$\delta = \text{degree}$

⑥ Determine whether the following graph G has an Euler cycle.

If the graph G has an Euler cycle, exhibit one.

(i)



$$\begin{aligned} \delta(v_1) &= 4, \delta(v_2) = 3, \delta(v_3) = 2 \\ \delta(v_4) &= 3, \delta(v_5) = 2, \delta(v_6) = 3 \\ \delta(v_7) &= 3, \delta(v_8) = 2, \delta(v_9) = \end{aligned}$$

Since the degree of vertices

v_1, v_2 and v_6 and v_7 are odd degree.
So, G has not an Euler cycle.

(iii) Six edges, eight vertices

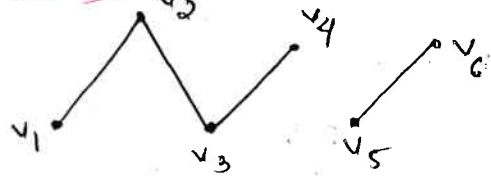
For any connected tree with v vertices, the number of edge e

$$e = v - 1.$$

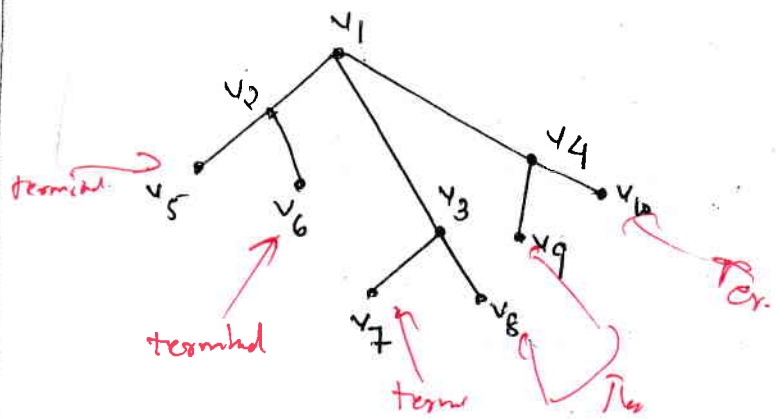
For $v = 8$, a connected tree requires $e = 8 - 1 = 7$ edges.

So, a connected tree does not exist.

(vi) Acyclic, Four edges; six vertices



(v) Tree; Four internal vertices; six terminal vertices



as graph is a tree

$v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8, v_9, v_{10}, v_{11}$